

# MGT 924: Statistical Foundations

## Problem Set 1

[Your Name]

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### 1 Visualize before building a prediction rule (20 points)

#### 1.1 Sample Statistics (1.1)

For each of the four datasets (I, II, III, IV), we computed the following statistics: the mean of  $\mathbf{x}$ , the mean of  $\mathbf{y}$ , the variance of  $\mathbf{x}$ , the variance of  $\mathbf{y}$ , the correlation between  $\mathbf{x}$  and  $\mathbf{y}$ , and the  $R^2$  implied by these statistics (using the identity  $R^2 = r_{xy}^2$  for simple linear regression).

Table 1: Sample statistics by dataset

| Dataset | Mean $x$ | Mean $y$ | Var $x$ | Var $y$ | Corr( $x, y$ ) | $R^2$  |
|---------|----------|----------|---------|---------|----------------|--------|
| I       | 9        | 7.5009   | 11      | 4.1273  | 0.8164         | 0.6665 |
| II      | 9        | 7.5009   | 11      | 4.1276  | 0.8162         | 0.6662 |
| III     | 9        | 7.5000   | 11      | 4.1226  | 0.8163         | 0.6663 |
| IV      | 9        | 7.5009   | 11      | 4.1232  | 0.8165         | 0.6667 |

#### 1.2 OLS Estimates (1.2)

For each dataset, we computed OLS estimates  $\hat{\beta}_0$  and  $\hat{\beta}_1$  from a regression of  $\mathbf{y}$  on  $\mathbf{x}$  using `lm()`.

Table 2: OLS estimates by dataset

| Dataset | $\hat{\beta}_0$ (Intercept) | $\hat{\beta}_1$ (Slope) |
|---------|-----------------------------|-------------------------|
| I       | 3.0001                      | 0.5001                  |
| II      | 3.0009                      | 0.5000                  |
| III     | 3.0025                      | 0.4997                  |
| IV      | 3.0017                      | 0.4999                  |

#### 1.3 Trading Rule from Statistics Alone (1.3)

*Based solely on the statistics computed in 1.1 and 1.2, how would you trade each of the four stocks?*

**Answer:**

From the statistics, we could observe a linear relationship between  $x$  and  $y$  approximately the same across all four stocks as

$$\hat{y} = 0.5x + 3,$$

with a positive correlation of approximately 0.816 and an  $R^2$  of approximately 0.666. Therefore, I would enter long positions in all four stocks when  $x$  is high and short positions when  $x$  is low.

## 1.4 Scatterplots with OLS Prediction Rule (1.4)

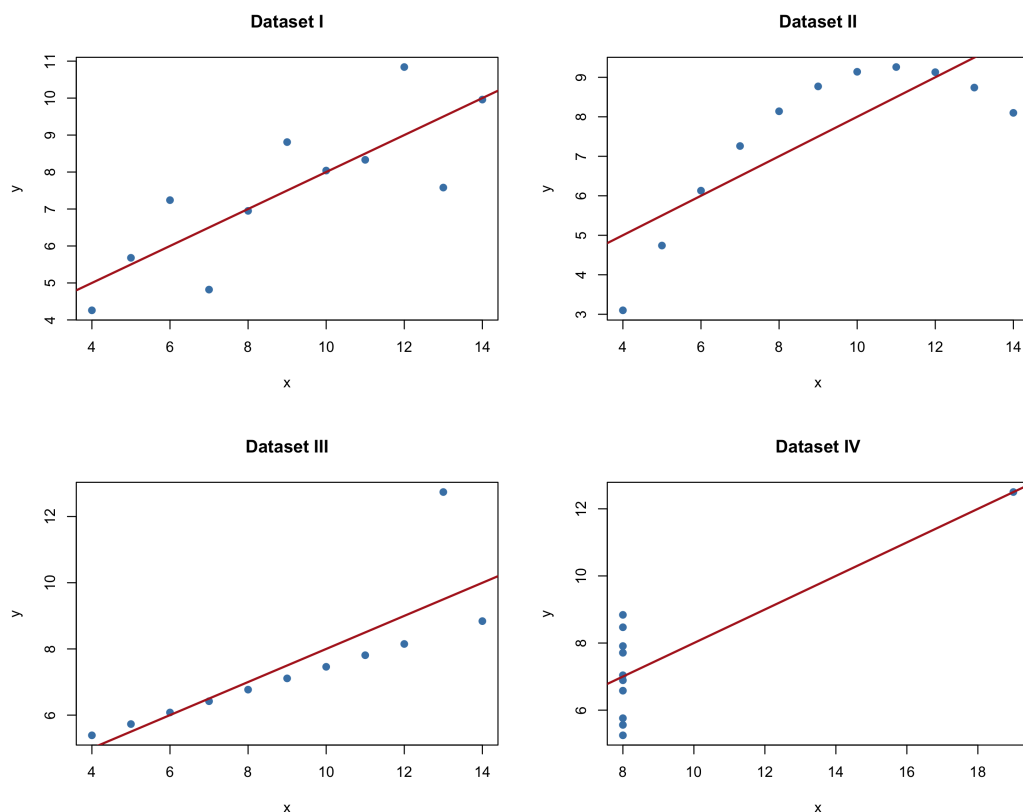


Figure 1: Scatterplots of  $y$  vs.  $x$  for each dataset, with the fitted OLS line overlaid.

## 1.5 Improving the Trading Strategy (1.5)

For each stock, describe the one improvement to your trading strategy (relative to your answer in 1.3) that the scatterplot suggests is most important.

**Answer:**

Although the four stocks have nearly identical summary statistics and OLS estimates, their scatterplots reveal very different patterns in the relationship between  $x$  and  $y$ . Therefore, the linear trading strategy we proposed in 1.3 should be adjusted separately for each stock based on the structure revealed by the scatterplots.

### Dataset I:

The data points are distributed roughly symmetrically around the fitted line. This suggests the linear relationship provides a reasonable description of the data. Therefore, I would retain the original strategy from 1.3: go long when  $x$  is high and short when  $x$  is low.

### Dataset II:

The scatterplot shows a clear nonlinear, inverted-U-shaped relationship of  $y = ax^2 + b$ .  $y$  initially increases with  $x$ , but after reaching a peak, the relationship becomes negative. Therefore, I would use different trading rules on the two sides of the turning point: go long as  $x$  rises before the peak, but reverse the strategy after the peak and short as  $x$  rises further.

### Dataset III:

The scatterplot shows that there is one unusual observation that pulls the entire fitted line upward, making the positive relationship between  $x$  and  $y$  appear stronger than it is for most of the data. Therefore, I would lower my expectation to the marginal returns. I would still go long when  $x$  is high and short when  $x$  is low, but with smaller positions and less confidence in the signal.

### Dataset IV:

In this plot, for most observations, there is no clear relationship between  $x$  and  $y$ . The apparent positive linear relationship is driven almost entirely by a single high-leverage observation.

Therefore, I would not rely on the fitted linear rule. And I would avoid taking long or short positions based on  $x$  alone without additional evidence.

## 2 Predicting house prices (40 points)

### 2.1 Data Preparation (2.1)

The following code loads the Connecticut property data, first constructs the true assessed value, then expresses both assessed value and sale amount in \$1,000s. Eventually we filter the sample to single-family homes in New Haven listed in 2008 with true assessed values below \$250,000.

```
# ---- 2.1.1: Load the data ----
ct <- read.csv("ps1/ct_property_data.csv")

# ---- 2.1.2: Create assessedvalue_true ----
# Tax-assessed value in CT is 70% of true market value,
# so dividing by 0.7 recovers the true value.
ct$assessedvalue_true <- ct$assessedvalue / 0.7

# ---- 2.1.3: Express in $1,000s ----
ct$assessedvalue_true <- ct$assessedvalue_true / 1000
ct$saleamount <- ct$saleamount / 1000

# ---- 2.1.4: Filter ----
ct_filtered <- ct[
  ct$residentialtype == "Single Family" &
```

```
ct$town == "New Haven" &  
ct$listyear == 2008 &  
ct$assessedvalue_true < 250,  
]
```

The resulting sample contains **279 observations**.